App Development Project Report

App Dev2 Project Report

1. Student Details

Name: Pushkar C
Roll Number: 24F2000951
Email:24f2000951@ds.study.iitm.ac.in
About Me: I am a Dual Degree Student pursuing BE CSE at BMSCE along with this online degree. I'm a person of curiosity,dedication who likes to push beyond boundaries and explore,experiment with new things. My interests are

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2. Project Details

Project Title: Placement Portal V2

Problem Statement:

Institutes require efficient systems to manage campus recruitment activities involving companies and students. Currently, many institutes rely on spreadsheets, emails, or manual coordination, which makes it difficult to manage company approvals, placement drives, student registrations, and application tracking.

Approach:

The portal was developed using a modern decoupled architecture, with **Vue.js** for the frontend UI and **Flask** for the backend API framework.It integrates **Celery** and **Redis** for asynchronous task management (such as periodic email alerts and monthly reports), **Chart.js** for interactive data visualizations and statistics, and **Mailpit** for local email testing.

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3. AI/LLM Declaration

Gemini was used as a supporting tool for structuring REST API endpoints,styling the webpages and improving naming consistency for better code readability.The AI/LLM usage is estimated at around 35-40%, primarily for code suggestions and minor refinements, without using any

external production APIs. All implementation, debugging, Vue component integration, and system setup were carried out independently

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4. Technologies and Frameworks Used

Technology/Library	Purpose
Vue js	Frontend Framework
Flask	Core backend web framework
Flask-SQLAlchmey	ORM for database management
Celery	Asynchronous task queue
Redis	In-memory message broker and cache
Chart.js	Analytics chart rendering
Mailpit	Local email testing tool
Flask-JWT-Extended	JWT authentication
SQLite	Lightweight local database
Werkzeug	Password Hashing

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5. Database Schema / ER Diagram

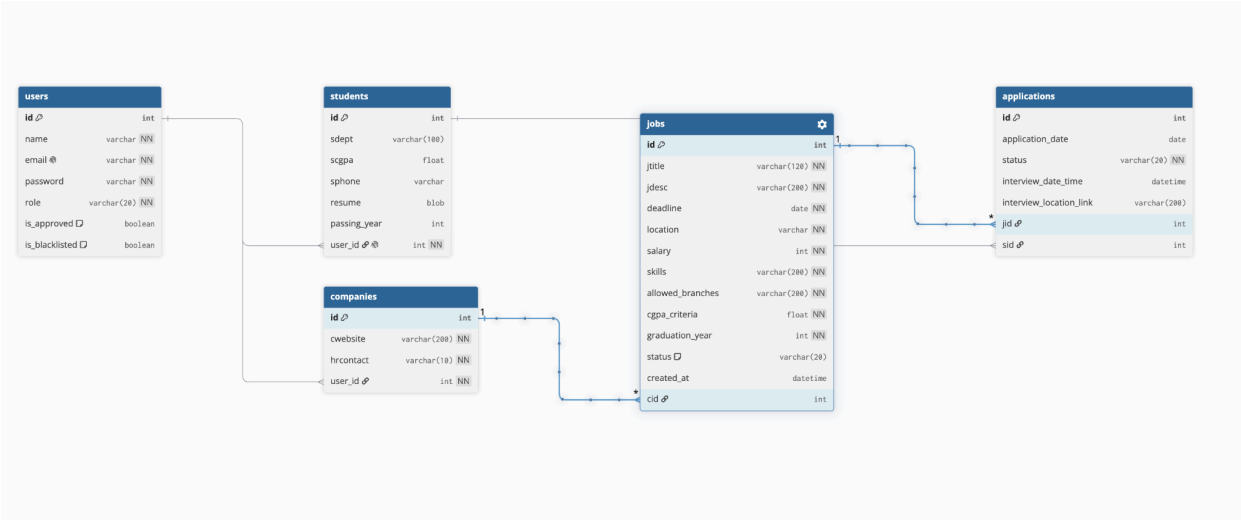
Tables:

1. **User** — stores user account details such as id, name, email, password, role, approval status, and blacklist status.
2. **Student** — stores student-specific information including id, department, CGPA, phone number, resume, passing year, and the associated user_id.
3. **Company** — stores company-specific details such as id, website, HR contact, and the associated user_id.

4. **Job** — stores job postings including id, title, description, deadline, location, salary, required skills, allowed branches, CGPA criteria, graduation year, status, creation timestamp, and the associated company_id.
5. **Application** — stores student job applications, including id, application date, application status, interview date and time, interview location or meeting link, and references to the associated job and student.

Relationships:

- **User ↔ Student:** One-to-One relationship. Each user with the **Student** role has one student profile, and each student profile belongs to exactly one user.
- **User ↔ Company:** One-to-One relationship. Each user with the **Company** role has one company profile, and each company profile belongs to exactly one user.
- **Company ↔ Job:** One-to-Many relationship. A company can post multiple jobs, while each job is posted by a single company.
- **Student ↔ Application:** One-to-Many relationship. A student can submit multiple job applications, while each application belongs to a single student.
- **Job ↔ Application:** One-to-Many relationship. A job can receive multiple applications, while each application is associated with a single job.
- **Student ↔ Job:** Many-to-Many relationship (implemented through the **Application** table). A student can apply to multiple jobs, and a job can receive applications from multiple students.



6. API Resource Endpoints

#	Method	Endpoint	Description
1	POST	/api/register	Register a new student or company
2	POST	/api/login	Authenticate user and obtain JWT token
3	GET	/api/admin/	Retrieve admin dashboard statistics
4	POST	/api/admin/companies/{id}/approve	Approve a company registration
5	POST	/api/admin/students/{id}/approve	Approve a student registration
6	POST	/api/admin/drives/{id}/approve	Approve a placement drive
7	GET / PUT	/api/company/profile	View or update company profile
8	POST	/api/company/create-drive	Create a new placement drive
9	GET	/api/company/drives/{id}/applicants	View applicants for a drive

10	**POST**	**/api/company/applicant/{id}/schedule**	**Schedule an interview for an applicant**
11	**GET / PUT**	**/api/student/profile**	**View or update student profile**
12	**GET**	**/api/student/drives**	**Retrieve available placement drives**
13	**POST**	**/api/student/drives/{id}/apply**	**Apply for a placement drive**
14	**GET**	**/api/student/applications**	**View submitted applications**
15	**GET**	**/api/generate-applications-report**	**Export applications report as a CSV file**

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7. Video Presentation

Drive Link:
https://drive.google.com/file/d/1_6DMOVYA-IkjIiZxA6UZ-LhJeLHFLzal/view?usp=sharing

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